Week 8 Reference Sheet: For Loops

CSC-125-201 | Use this as a reference when completing Homework 8

Homework Topics

- Basic for loop syntax
- Iterating through lists
- Building new lists in loops
- The range() function (3 forms)
- Index-based loops with range(len())
- Looping through strings
- Accumulating totals
- Counting and filtering
- Parallel list processing
- Nested loops

Basic For Loop

Repeats code once for each item in a sequence. Uses colon and indentation.

```
for item in sequence:
    # runs once per item
# code here runs after loop finishes
```

The range() Function

Form	Produces	Example	Output
`range(stop)`	0 to stop - 1	`range(5)`	0, 1, 2, 3, 4
`range(start, stop)`	start to stop - 1	`range(1, 6)`	1, 2, 3, 4, 5
`range(start, stop, step)`	start to stop - 1 by step	`range(0, 10, 2)`	0, 2, 4, 6, 8
`range(start, stop, step)`	Negative step counts down	`range(5, 0, -1)`	5, 4, 3, 2, 1

Stops **before** the stop number. To include 5, use `range(1, 6)`.

Direct Iteration vs Index-Based

Approach	When to use	Syntax
Direct	You just need each value	`for item in my_list:`
Index-based	Need position, modification, or parallel lists	`for i in range(len(my_list)):`

Common Loop Patterns

```
# Accumulate a total
total = 0
for price in prices:
    total = total + price

# Count matches
count = 0
for score in scores:
    if score ≥ 90:
        count = count + 1

# Filter into a new list
result = []
for num in numbers:
    if num > 10:
        result.append(num)

# Build a transformed list
doubled = []
for num in numbers:
    doubled.append(num * 2)
```

Finding Max/Min with Position

```
best = scores[0]
best_index = 0
for i in range(len(scores)):
    if scores[i] > best:
        best = scores[i]
        best_index = i
```

Parallel Lists

Two lists share the same indexes, so `names[i]` and `scores[i]` refer to the same person.

```python
names = ["Alice", "Bob"]
scores = [85, 92]
for i in range(len(names)):
    print(f"{names[i]}: {scores[i]}")
```

String Iteration

```python
# Count vowels
count = 0
for char in text:
    if char.lower() in "aeiou":
        count = count + 1

# Build new string (filter characters)
result = ""
for char in text:
    if char.isalpha():
        result = result + char
```

Nested Loops

Inner loop runs fully for each outer iteration. 3 outer x 4 inner = 12 runs.

```python
for row in range(3):
    for col in range(4):
        print(f"({row},{col})", end=" ")
    print()  # new line after each row
```

Common Errors

Error	Cause	Fix
`IndentationError`	Loop body not indented	Indent 4 spaces after `for`
Off-by-one	`range(1, 5)` excludes 5	Use `range(1, 6)` to include 5
Skipped items	Modifying list while looping	Build a new list instead
Wrong print placement	Print inside loop instead of after	Check indentation level
`IndexError`	`range(len(list) + 1)`	Use `range(len(list))`

Resources

- W3Schools - Python For Loops
- LearnPython.org - Interactive Loop Tutorial
- Real Python - For Loops Guide
- W3Schools - range() Function